

Exercise 16. Deep boat

1) The state equations are:

$$\begin{cases} \dot{x} = v \cos \theta \\ \dot{y} = v \sin \theta \\ \dot{\theta} = u_1 - u_2 \\ \dot{v} = u_1 + u_2 \end{cases}$$

with u_1, u_2 the propellers controlled at each side of the boat.

The given controller is something like $u = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \beta(v) \\ g(y, \theta) \end{pmatrix}$

Let's see v, θ :

$$\begin{pmatrix} \dot{v} \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} u_1 + u_2 \\ u_1 - u_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}}_u$$

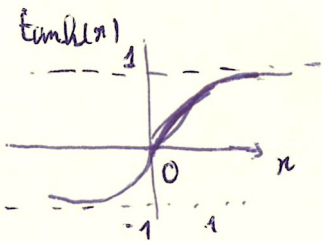
An interesting controller could be something like $u = A^{-1} \cdot v$

$$A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

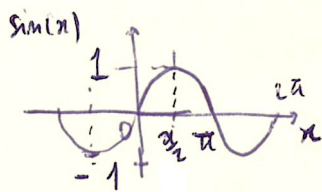
Regarding the given controller, we will have something like:

$$\begin{pmatrix} \dot{v} \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} u_1 + u_2 \\ u_1 - u_2 \end{pmatrix} = \underbrace{C}_\text{constant} \begin{pmatrix} \beta(v) \\ g(y, \theta) \end{pmatrix} = C \cdot \begin{pmatrix} \tanh(1-v) \\ \sin(-\tanh(\frac{y}{10}) - \theta) \end{pmatrix}$$

constant

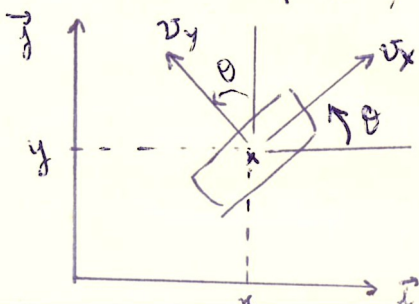


So the boat will go forward if v is less than the desired 1, because $\tanh(1-v) > 0$ then $\dot{v} = u_1 + u_2 > 0$. A positive acceleration.



Having the $\sin(\Delta\theta)$ will help the boat to turn quickly when $\Delta\theta \neq 0$ with the sign difference of $u_1 - u_2$.

2) The new client boat slides sideways when changing its heading. So we have another speed v_y that we have to take into account:



we can conclude:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = R_\theta \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix}$$